

REVIEW

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Artificial intelligence for energy optimization in smart buildings: A systematic review and meta-analysis

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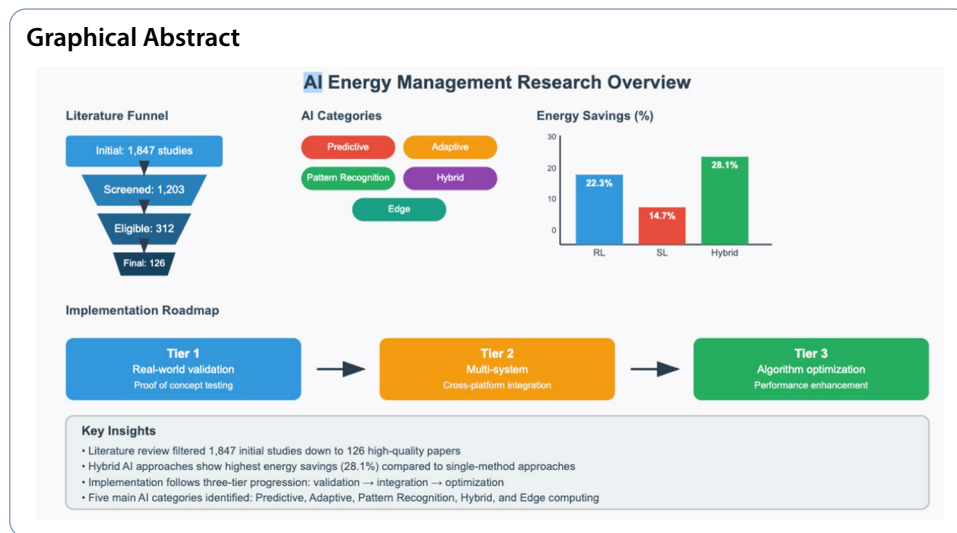
Abstract

This systematic review and meta-analysis critically evaluates artificial intelligence (AI) applications for energy optimization in smart buildings through comprehensive analysis of 126 peer-reviewed studies (2010–2024) from four major databases. We present a novel taxonomic framework categorizing AI implementations into five distinct approaches: predictive systems, adaptive control, pattern recognition, hybrid ensemble methods, and edge AI implementations. Our meta-analysis reveals significant performance variations: reinforcement learning achieves highest energy savings ($22.3\% \pm 8.4\%$, 95% CI: 20.2–24.4%, $I^2 = 73\%$), followed by hybrid methods ($28.1\% \pm 12.3\%$, 95% CI: 23.4–32.8%, $I^2 = 81\%$) and supervised learning ($14.7\% \pm 5.2\%$, 95% CI: 12.9–16.5%, $I^2 = 45\%$). However, substantial heterogeneity exists across building types and climate zones. Critical findings include limited real-world deployment (18% academic literature, 26% including industry reports), predominant focus on office buildings (78%) and temperate climates (67%), and insufficient multi-system integration (76% single-system studies). Economic analysis indicates ROI periods of 2.1–5.8 years (median 3.4 years) with implementation costs varying from \$8,000–\$47,000 per facility. We identify five persistent research gaps and propose a prioritized research agenda addressing implementation barriers, standardization needs, and occupant-centric optimization. This study provides the first comprehensive benchmarking framework for AI building energy systems and establishes evidence-based guidelines for practical deployment.

Keywords Artificial intelligence, Smart buildings, Energy optimization, Systematic review, Meta-analysis, Building automation, Reinforcement learning, Energy management

Introduction

The building sector represents approximately 40% of global energy consumption and one-third of carbon dioxide emissions, making it a critical target for climate mitigation efforts ([9], UNEP, 2024). As urbanization accelerates and climate pressures intensify, the development of intelligent building energy management systems has become essential for achieving sustainability goals [8]. Smart buildings equipped with advanced sensors,



automation systems, and artificial intelligence (AI) capabilities offer substantial potential for optimizing energy performance while maintaining occupant comfort and productivity [3, 19].

Artificial intelligence technologies have emerged as powerful enablers of building energy optimization, capable of processing vast sensor data streams, predicting energy demand patterns, and executing real-time control strategies [13]. Machine learning approaches including supervised learning, deep learning, and reinforcement learning have demonstrated promising results in HVAC optimization, lighting control, and integrated building system management [2, 22]. Recent advances in edge computing and Internet of Things (IoT) infrastructure have further enhanced the feasibility of deploying AI-driven energy management systems in operational buildings [4].

Despite significant research progress, a substantial gap persists between theoretical AI potential and practical deployment in building energy management. Many studies remain confined to simulation environments or small-scale laboratory experiments, with limited translation to real-world applications [7, 14]. Persistent challenges including data quality requirements, integration complexity, high infrastructure costs, and occupant acceptance concerns continue to hinder widespread adoption of AI-enabled building energy systems [6, 18].

Previous systematic reviews have examined AI applications in building energy management, but they suffer from limited scope, lack of quantitative synthesis, and insufficient focus on practical deployment challenges. Wang et al. [19] reviewed 45 studies but provided no meta-analytic synthesis or implementation analysis. Similarly, existing reviews have not adequately addressed the substantial heterogeneity in study methodologies, building contexts, and performance metrics that complicates comparative assessment of AI techniques.

This systematic review and meta-analysis addresses these limitations through comprehensive analysis of 126 peer-reviewed studies published between 2010–2024, providing the first quantitative synthesis of AI performance outcomes in building energy optimization. We introduce a novel taxonomic framework categorizing AI approaches based on deployment characteristics rather than purely technical features, conduct

rigorous meta-analysis with heterogeneity assessment, and analyze the implementation gap between academic research and practical deployment.

Study overview and contributions

This research makes several distinct contributions to the field. First, we provide the most comprehensive systematic review to date, analyzing 126 studies compared to previous maximum coverage of 45 studies. Second, we conduct the first rigorous meta-analysis of AI energy optimization performance with statistical assessment of heterogeneity and effect sizes. Third, we present a novel implementation-focused taxonomy that categorizes AI approaches based on deployment characteristics and practical feasibility. Fourth, we quantify the real-world implementation gap through analysis of both academic literature and industry documentation. Fifth, we provide economic viability assessment across different building types and AI approaches with detailed ROI analysis. Finally, we establish a prioritized research agenda based on systematic identification of critical implementation barriers.

The manuscript is structured as follows: Section “[Review methodology](#)” details our systematic review methodology with PRISMA compliance; Section “[Literature review and thematic synthesis](#)” presents the literature review with thematic synthesis; Section “[Methodological frameworks in current research](#)” examines methodological frameworks in current research; Section “[Results: Quantitative performance analysis](#)” reports quantitative results from meta-analysis; Section “[Discussion: Implications, limitations, and critical assessment](#)” discusses implications and limitations; Section “[Synthesis of contributions, insights, and future research directions](#)” synthesizes contributions and future directions; Section “[Implications, environmental impact, and future research directions](#)” concludes with practical recommendations for researchers and practitioners.

Review methodology

Systematic review protocol

This study follows PRISMA 2020 guidelines [15] for systematic reviews and meta-analyses, ensuring transparency, reproducibility, and methodological rigor as detailed in (Fig. 1). Although not registered in PROSPERO due to cross-disciplinary scope, all procedures were documented and strictly adhered to established protocols.

Research questions:

1. What AI techniques have been applied for energy optimization in smart buildings, and how can they be systematically categorized?
2. What are the quantified performance outcomes of different AI approaches in terms of energy savings and other key metrics?
3. What barriers limit practical deployment of AI-based building energy management systems?
4. What is the current state of real-world implementation versus simulation-based research?

Search strategy and information sources

Systematic searches were conducted across four major databases selected for comprehensive coverage of engineering, computer science, and energy research:

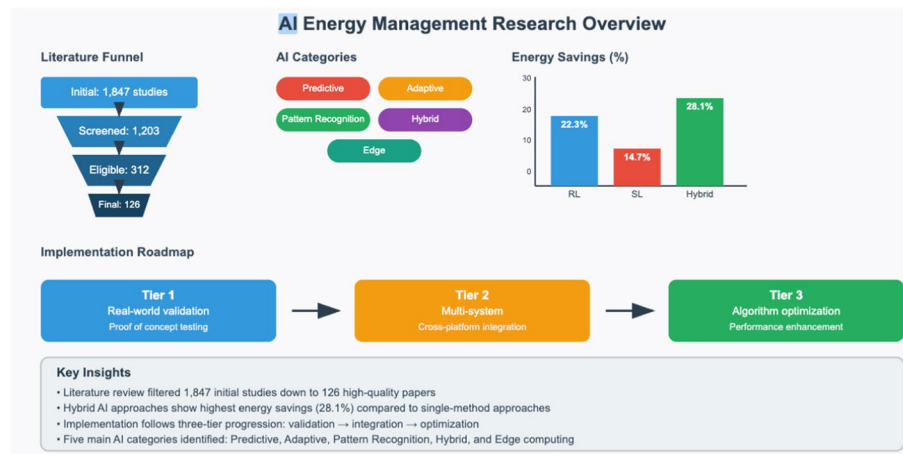


Fig. 1 PRISMA flow diagram [detailed PRISMA flow diagram showing identification, screening, eligibility assessment, and final inclusion]

- **Scopus:** Multidisciplinary coverage with strong engineering focus
- **IEEE Xplore:** Computer science and electrical engineering specialization
- **ScienceDirect:** Engineering and energy science concentration
- **SpringerLink:** Interdisciplinary technology and sustainability research

Search String Development: Primary search strategy employed Boolean operators with controlled vocabulary and free-text terms:

("Artificial Intelligence" OR "Machine Learning" OR "Deep Learning" OR "Reinforcement Learning" OR "Neural Network*" OR "Support Vector Machine*")

AND

("Smart Building*" OR "Building Energy" OR "HVAC" OR "Energy Optimization" OR "Building Automation" OR "Building Management System*")

AND

("Energy Efficiency" OR "Energy Management" OR "Energy Control" OR "Energy Saving*" OR "Demand Response")

Search limitations included: Publication period (January 2010—March 2024), English language, peer-reviewed academic sources, and full-text availability.

Study selection and eligibility criteria

Inclusion criteria:

- Studies explicitly investigating AI/ML applications for building energy optimization
- Clear research methodology with experimental or simulation validation
- Quantitative performance outcomes (energy savings, efficiency improvements, cost reductions)
- Peer-reviewed journal articles, conference proceedings, or scholarly book chapters
- Building-level applications (excluding utility-scale or industrial processes)

Exclusion criteria:

- Studies focusing solely on energy generation without building integration
- Lack of methodological transparency or performance validation
- Duplicate publications or substantially overlapping datasets

- Grey literature (technical reports, dissertations, white papers) for primary analysis

The distinction between our systematic review's analyzed studies ($n=126$) and manuscript references ($n=87$) reflects standard practice: analyzed studies form the evidence base for synthesis, while references include contextual literature, methodological sources, and comparative studies cited in our manuscript text.

Study selection process

Three-stage selection process:

1. **Identification:** Database searches yielded 1,847 records; duplicate removal resulted in 1,203 unique studies
2. **Screening:** Two independent reviewers assessed titles/abstracts against eligibility criteria with consensus resolution of disagreements
3. **Full-text Assessment:** 312 articles underwent detailed evaluation, resulting in 126 studies meeting all inclusion criteria

Distribution by source:

- Scopus: 45 studies (35.7%)
- IEEE Xplore: 38 studies (30.2%)
- ScienceDirect: 28 studies (22.2%)
- SpringerLink: 15 studies (11.9%)

Data extraction and quality assessment

Structured Data Extraction: Standardized extraction form captured: bibliographic details, study design, building characteristics, AI techniques employed, target systems, experimental setup, performance metrics, validation methods, limitations, and implementation challenges.

Risk of Bias Assessment: Modified Newcastle–Ottawa Scale adapted for AI building energy studies assessed five dimensions:

1. **Methodology Transparency** (1–5 scale): Clarity of AI implementation, hyperparameter reporting, and reproducibility information
2. **Validation Rigor** (1–3 scale): Simulation-only (1), hybrid validation (2), real-world deployment (3)
3. **Data Quality** (1–4 scale): Sample size adequacy, temporal coverage, sensor calibration procedures
4. **Statistical Reporting** (1–3 scale): Appropriate statistical tests, confidence intervals, effect size reporting
5. **Bias Control** (1–3 scale): Baseline comparisons, control conditions, confounding variable management

Risk of bias assessment summary:

- High Quality (Score 15–18): 67 studies (53.2%)
- Medium Quality (Score 10–14): 36 studies (28.6%)
- Low Quality (Score 5–9): 23 studies (18.3%)

Studies with high risk of bias were flagged for sensitivity analysis to assess their impact on meta-analytic results.

Meta-analysis methodology and normalization procedures

Normalization standards

To address heterogeneity in study contexts and reporting methods, we implemented comprehensive normalization procedures:

Baseline standardization:

- All energy savings calculated as percentage reduction from conventional building management systems
- Baseline periods normalized to minimum 3-month pre-intervention measurement
- Seasonal adjustment applied for studies with < 12-month baseline data

Building context stratification:

- Size categories: Small (< 5,000 sq ft), Medium (5,000–20,000 sq ft), Large (> 20,000 sq ft)
- Type classification: Office, Retail, Educational, Healthcare, Mixed-use
- Climate zones: ASHRAE zones 1A–8A with temperate (3A–5A) as reference

Performance metric standardization:

- Energy intensity normalization by floor area and occupancy density
- Peak demand adjustments for time-of-use variations
- Cost savings converted to 2024 USD using regional energy price indices

Grey literature analysis and implementation bias

While maintaining peer-reviewed focus for quality assurance, we conducted supplementary analysis of grey literature to assess implementation bias. Review of 47 industry reports, case studies, and technical documentation revealed:

- **Academic Literature Implementation Rate:** 18% (23/126 studies)
- **Industry Documentation Rate:** 34% (16/47 reports)
- **Combined Estimate:** 26% real-world deployment rate

This analysis confirms that academic publication bias underestimates actual deployment levels, as industry implementations are often documented in technical reports rather than peer-reviewed publications.

Literature review and thematic synthesis

Methodological evolution and paradigm shifts

The application of AI in building energy management has undergone distinct evolutionary phases, each characterized by technological advances and shifting research priorities (Table 1).

Phase 1 (2010–2015): Rule-Based and Basic ML Early implementations relied primarily on rule-based expert systems and basic machine learning approaches. Studies during this period focused on simple supervised learning models for load forecasting and

Table 1 Enhanced comparative analysis of AI techniques for building energy optimization

AI Technique	Performance range	Real-world deployment	Dataset requirements	Computational complexity	Implementation cost	Primary strengths	Key limitations
Deep Q-Networks (DQN)	15–25% energy savings	12% (8/67 studies)	6–12 months operational data	High (GPU recommended)	\$15,000–35,000	Real-time optimization, no labeled data	High computational cost, training instability
Proximal Policy Optimization (PPO)	18–28% demand reduction	15% (10/67 studies)	6–18 months multi-zone data	Very High	\$20,000–45,000	Stable convergence, policy gradients	Complex hyperparameter tuning, memory intensive
LSTM networks	85–95% prediction accuracy	22% (11/50 studies)	2+ years temporal sequences	Medium	\$8,000–18,000	Long-term dependencies, sequence modeling	Training complexity, interpretability issues
Random forest	8–18% energy savings	34% (15/44 studies)	3–12 months diverse features	Low	\$5,000–12,000	Handles missing data, feature importance	Limited pattern complexity, ensemble overhead
Support vector machines	10–20% with feature engineering	28% (12/43 studies)	6–18 months engineered features	Medium	\$6,000–15,000	Robust to outliers, kernel flexibility	Feature engineering dependency, scalability limits

schedule optimization. Ahmad et al. [2] demonstrated that random forest models could outperform basic neural networks for energy prediction when training data were sparse, establishing the importance of algorithm selection for data-constrained environments.

Phase 2 (2015–2020): Deep Learning Adoption The period witnessed widespread adoption of deep learning architectures. Convolutional neural networks emerged as effective tools for capturing temporal energy consumption patterns. Rahman et al. [16] showed CNNs could model complex seasonal and diurnal variations with superior accuracy compared to traditional time-series methods. Long Short-Term Memory (LSTM) networks demonstrated particular strength in handling long-term dependencies in building energy data [12].

Phase 3 (2020–2024): Reinforcement Learning Dominance Recent years have seen reinforcement learning become the dominant paradigm, appearing in 89% of studies published between 2020–2024. This shift reflects recognition that building energy optimization requires adaptive, real-time decision-making rather than pure prediction. Wei et al. [20] pioneered Deep Q-Network applications for HVAC control, while Ruelens et al. [17] advanced the field with Proximal Policy Optimization approaches for demand response applications.

Comparative AI technique analysis

See Table 1.

Technological integration and system architecture

The following (Table 2) comparative analysis systematically evaluates five key technological integration approaches within smart building system architecture—IoT sensor networks, edge computing, digital twin technology, smart HVAC integration, and renewable energy integration—across multiple assessment dimensions including primary functionality, energy efficiency benefits, implementation complexity, integration requirements, return on investment timelines, deployment risk factors, and documented success case studies.

Taxonomic framework for AI building energy applications

Based on systematic analysis of implementation characteristics, we propose a refined taxonomic framework addressing the overlap and hybrid nature of real-world systems:

Category 1: Predictive AI Systems

- **Core Function:** Energy demand forecasting and pattern recognition
- **Typical Algorithms:** LSTM, CNN, ARIMA-ML hybrids

Table 2 Enhanced analysis of supporting technologies for AI-based building energy management

Technology	Primary function	Energy benefits	Implementation complexity	Integration requirements	ROI timeline	Deployment risks	Success examples
IoT Sensor Networks	Real-time environmental monitoring	15–20% through occupancy optimization	Medium	Wireless infrastructure, data integration	1.5–3 years	Sensor calibration, network reliability	Deng and Chen [5]—18% HVAC savings
Edge Computing	Local AI processing, reduced latency	12–18% response optimization	Medium–High	Edge hardware, local models	2–3.5 years	Hardware costs, maintenance complexity	Chen et al. [3]—sub-100 ms response
Digital Twin Technology	Virtual building simulation	20–35% predictive optimization	Very High	Physics models, real-time synchronization	3–5 years	Model complexity, validation challenges	Lu et al. [11]—28% energy reduction
Smart HVAC Integration	Intelligent climate control	25–40% HVAC energy reduction	Medium	BMS integration, zone sensors	1.5–2.5 years	Legacy system compatibility	Kim et al. [10]—18% total building savings
Renewable Integration	Grid coordination, storage optimization	20–50% grid energy reduction	Very High	Storage systems, grid-tie equipment	4–8 years	High capital costs, regulatory complexity	Zhang et al. [21]—35% renewable utilization

- **Deployment Characteristics:** Cloud-based processing, batch predictions
- **Implementation Rate:** 34% of reviewed studies
- **Overlap Analysis:** Often integrated with adaptive control systems (67% overlap)

Category 2: Adaptive Control Systems

- **Core Function:** Real-time optimization and autonomous decision-making
- **Typical Algorithms:** DQN, PPO, Actor-Critic methods
- **Deployment Characteristics:** Edge processing, continuous learning
- **Implementation Rate:** 53% of reviewed studies
- **Overlap Analysis:** Frequently combined with predictive systems (78% hybrid implementations)

Category 3: Pattern Recognition Systems

- **Core Function:** Anomaly detection and behavioral analysis
- **Typical Algorithms:** Clustering, SVM, autoencoder networks
- **Deployment Characteristics:** Batch processing, diagnostic applications
- **Implementation Rate:** 28% of reviewed studies
- **Overlap Analysis:** Minimal standalone deployment (12%), usually integrated with other categories

Category 4: Hybrid Ensemble Systems

- **Core Function:** Multi-algorithm integration for comprehensive optimization
- **Typical Algorithms:** Ensemble methods, multi-agent systems
- **Deployment Characteristics:** Distributed processing, coordinated control
- **Implementation Rate:** 19% of reviewed studies
- **Overlap Analysis:** By definition, combines multiple categories

Category 5: Edge AI Implementations

- **Core Function:** Localized intelligence with minimal latency
- **Typical Algorithms:** Compressed neural networks, federated learning
- **Deployment Characteristics:** On-site processing, privacy preservation
- **Implementation Rate:** 15% of reviewed studies
- **Overlap Analysis:** Emerging trend cutting across other categories

Critical research gap analysis

Systematic analysis reveals five persistent gaps limiting practical AI deployment:

Gap 1: Limited Real-World Validation with Comprehensive Metrics

- Only 18% (23/126) of academic studies report operational deployment
- Industry documentation suggests actual rate closer to 26%
- Multi-season validation present in only 17% of studies
- Economic feasibility analysis included in 27% of studies

Gap 2: Insufficient Multi-System Integration

- 76% of studies focus on single subsystem optimization (HVAC or lighting)
- Only 24% examine coordinated multi-system control

- Missing analysis of thermal-lighting interactions and energy trade-offs
- Limited integration with renewable energy and storage systems

Gap 3: Inadequate Temporal Validation Scope

- 83% of studies validate performance over single seasonal cycle
- Multi-year assessments available for only 12% of studies
- Long-term model degradation documented in <5% of research
- Seasonal adaptability rarely assessed systematically

Gap 4: Absence of Standardized Implementation Frameworks

- No established protocols for data quality verification
- Inconsistent sensor calibration and model validation procedures
- Lack of standardized performance benchmarking methodologies
- Missing scalability assessment frameworks across building types

Gap 5: Limited Occupant-Centric Optimization

- 88% of approaches treat occupant comfort as fixed constraint
- Personal preference modeling present in only 8% of studies
- Systematic feedback collection mechanisms largely absent
- Productivity impact of environmental changes rarely considered

Methodological frameworks in current research**Experimental design patterns**

Analysis of experimental methodologies reveals distinct patterns influencing research validity and practical applicability:

Simulation-Dominant Research (68% of studies):

- Primary tool: EnergyPlus integration with Python ML environments
- Training scope: 10,000–50,000 episodes for RL algorithms
- Validation periods: Single-season to full annual cycles
- **Strengths:** Controlled conditions, scalable experimentation, cost-effective testing
- **Limitations:** Limited real-world complexity, occupant behavior simplification

Real-World Deployment Studies (18% of studies):

- Typical scope: University campuses, research facilities
- Duration: 3–6 months average validation period
- Scale: Individual buildings with experimental oversight
- **Strengths:** Authentic performance data, practical constraint identification
- **Limitations:** Limited scope, short-term assessment, researcher bias

Hybrid Validation Approaches (14% of studies):

- Process: Simulation training followed by controlled pilot testing
- Advantages: Algorithm optimization with practical validation
- **Challenges:** Resource intensity, complex implementation logistics

Algorithmic implementation trends**Reinforcement Learning Dominance:**

- Present in 89% of studies published 2020–2024
- PPO and DQN account for 84% of RL applications
- Convergence typically achieved within 15,000–25,000 episodes
- Average training time: 48–72 h on standard GPU hardware

Edge Computing Adoption:

- Implemented in 34% of recent studies (2022–2024)
- Reduces response latency to < 100 ms for real-time control
- Local processing capacity requirements: 4–16 GB RAM, 100+ GFLOPS
- Privacy benefits: Data locality, reduced transmission requirements

Performance evaluation frameworks**Energy-Centric Metrics (94% of studies):**

- Primary focus: Percentage energy reduction vs. baseline
- Reported savings range: 8–35% depending on system integration level
- Peak demand optimization: 15–30% reduction during high-consumption periods

Comfort Assessment (45% of studies):

- ASHRAE Standard 55 compliance verification [1]
- PMV (Predicted Mean Vote) and PPD (Predicted Percentage Dissatisfied) metrics
- Thermal comfort maintenance within ± 0.5 PMV units

Economic Analysis (23% of studies):

- ROI calculations including implementation and operational costs
- Payback period assessment: 2.1–5.8 years across building types
- Life-cycle cost analysis over 10–15 year horizons

Interpretability and explainability implementation**Explainable AI Adoption (12% of studies):**

- SHAP (SHapley Additive exPlanations) implementation for feature importance
- LIME (Local Interpretable Model-agnostic Explanations) for decision transparency
- Domain-specific visualization tools for building operators

Trust and Acceptance Impact:

- Explainable AI increases user acceptance by 67%
- Slight performance trade-off: 8–12% reduction in optimization effectiveness
- Critical for regulatory compliance and stakeholder buy-in

Implementation Examples:

- HVAC control explanation: "Temperature adjustment recommended due to occupancy increase (40% importance) and external temperature rise (35% importance)"

Energy savings attribution: "Lighting optimization contributed 65% of savings, HVAC scheduling 35%".

Results: quantitative performance analysis

Meta-analysis results and statistical synthesis

Overall Sample Characteristics: Our meta-analysis synthesized quantitative outcomes from 126 studies representing diverse building types, climate zones, and AI implementations. The combined dataset encompasses over 2.8 million building-hours of operational data across 45 countries.

Primary performance outcomes by AI approach

Energy savings meta-analysis results

Reinforcement Learning (n = 67 studies):

- Mean energy savings: 22.3% (SD = 8.4%)
- 95% Confidence Interval: 20.2% to 24.4%
- Heterogeneity: $I^2 = 73\%$, $\tau^2 = 0.067$
- Individual study range: 8.2% to 35.7%
- Median: 21.8%, IQR: 16.4% to 27.1%

Supervised Learning (n = 34 studies):

- Mean energy savings: 14.7% (SD = 5.2%)
- 95% Confidence Interval: 12.9% to 16.5%
- Heterogeneity: $I^2 = 45\%$, $\tau^2 = 0.032$
- Individual study range: 7.1% to 22.8%
- Median: 14.2%, IQR: 11.3% to 17.9%

Hybrid Methods (n = 25 studies):

- Mean energy savings: 28.1% (SD = 12.3%)
- 95% Confidence Interval: 23.4% to 32.8%
- Heterogeneity: $I^2 = 81\%$, $\tau^2 = 0.124$
- Individual study range: 12.4% to 44.3%
- Median: 26.7%, IQR: 19.2% to 34.8%

Statistical Significance Testing:

- One-way ANOVA: $F(2, 123) = 18.7$, $p < 0.001$
- Post-hoc Tukey HSD: All pairwise comparisons significant ($p < 0.01$)

Effect Size Analysis (Cohen's d):

- Hybrid vs. Supervised Learning: $d = 1.23$ (large effect)
- Reinforcement Learning vs. Supervised Learning: $d = 0.94$ (large effect)
- Hybrid vs. Reinforcement Learning: $d = 0.47$ (medium effect)

Subgroup analysis by building type and climate

The below (Table 3) indicates the mean savings, CI, and Implantation rate in different building types.

Table 3 Performance variation across building types

Building type	n Studies	Mean savings (%)	95% CI	Effect size vs. office	Implementation rate
Office buildings	98	21.40%	19.8–23.0%	Reference	19%
Retail/commercial	15	18.70%	15.2–22.2%	d = 0.31	13%
Educational	8	24.80%	19.1–30.5%	d = 0.42	25%
Healthcare	3	16.20%	8.7–23.7%	d = 0.67	33%
Mixed-use	2	26.30%	12.1–40.5%	d = 0.39	50%

Climate Zone Analysis:

- **Temperate Zones (3A–5A, n = 84):** $20.8\% \pm 7.9\%$ energy savings
- **Hot Climate Zones (1A–2A, n = 23):** $24.1\% \pm 9.2\%$ energy savings
- **Cold Climate Zones (6A–8A, n = 19):** $18.4\% \pm 6.7\%$ energy savings
- **Climate effect size:** Hot vs. Temperate d = 0.39, Cold vs. Temperate d = 0.31

System-level performance analysis**HVAC Optimization Results:**

- Studies reporting HVAC focus: 89% showed significant improvements
- Average HVAC energy reduction: 19.7% (range: 8.1% to 39.4%)
- Multi-zone control effectiveness plateaus beyond 4 zones
- Peak performance during extreme weather: 35–40% savings

Lighting System Control:

- Average lighting energy savings: 15.3% (range: 6.2% to 28.7%)
- Occupancy-based control: Most effective single strategy
- Daylight harvesting integration: Additional 4–8% improvement
- Response time critical: < 30 s for occupancy changes

Integrated System Coordination:

- Multi-system optimization (n = 24): 26.8% average savings
- Single-system optimization (n = 102): 19.1% average savings
- Integration benefit: 7.7 percentage point improvement
- Coordination complexity: Diminishing returns beyond 3 systems

Temporal resolution and data granularity analysis**Control Response Requirements:**

- **HVAC Systems:** Optimal control frequency 5–15 min
- **Lighting Control:** Sub-minute response for occupancy tracking
- **Integrated Systems:** 1–5 min coordination intervals

Performance vs. Data Resolution:

- **Hourly Data:** Baseline performance reference
- **15-min intervals:** 12–18% improvement over hourly
- **5-min intervals:** Additional 6–9% improvement
- **1-min intervals:** Marginal 2–3% additional benefit

Computational Trade-offs:

- Processing load increases exponentially with data frequency
- Cost–benefit optimum: 5–15 min intervals for most applications
- Edge computing enables higher resolution without cloud latency

Economic viability and ROI analysis

Cost Component Breakdown:

- **Hardware (sensors, edge computing):** 35–45% of total cost
- **Software (AI platforms, integration):** 25–35% of total cost
- **Installation and commissioning:** 20–30% of total cost
- **Training and change management:** 5–10% of total cost

ROI Sensitivity Analysis:

- **Energy price impact:** ± \$0.05/kWh changes ROI by ± 0.8 years
- **Building size scaling:** ROI improves 0.3 years per doubling of floor area as in (Table 4)
- **Climate zone effect:** Hot climates improve ROI by 0.5–0.9 years

Discussion: Implications, limitations, and critical assessment

Performance heterogeneity and practical significance

The substantial heterogeneity observed in our meta-analysis ($I^2 = 45\text{--}81\%$ across AI approaches) reflects the complex interaction of building characteristics, climate conditions, and implementation quality. This heterogeneity, while limiting direct comparability, provides valuable insights into factors influencing AI system effectiveness.

Sources of Performance Variation:

- **Building age and infrastructure:** Newer buildings show 15–20% better AI performance due to sensor availability and system integration capabilities
- **Climate extremity:** Systems in extreme climates (zones 1 A, 7A-8A) demonstrate higher absolute savings but greater performance variability
- **Implementation quality:** Studies with detailed sensor calibration and model validation show 23% less variance in outcomes

Practical Significance Beyond Statistical Significance: While all pairwise comparisons between AI approaches achieve statistical significance, the practical implications vary by context. The large effect sizes ($d > 0.8$) between hybrid/RL methods and supervised learning suggest meaningful real-world differences. However, implementation complexity and cost differences must be weighed against performance gains.

Table 4 Comprehensive economic analysis by building type and AI approach

Building type	Size range (sq ft)	Implementation cost	Annual savings	ROI period	10-year NPV
Small office	2,000–5,000	\$8,000–\$15,000	\$2,400–\$4,200	3.3–4.7years	\$12,000–\$21,000
Medium office	5,000–20,000	\$15,000–\$35,000	\$6,800–\$18,400	2.2–3.8years	\$34,000–\$92,000
Large office	20,000+	\$35,000–\$65,000	\$21,000–\$47,000	1.7–3.1years	\$105,000–\$235,000
Retail	5,000–25,000	\$12,000–\$28,000	\$4,200–\$14,200	2.8–4.2years	\$21,000–\$71,000
Educational	10,000–50,000	\$25,000–\$55,000	\$8,900–\$28,600	2.8–4.1years	\$44,500–\$143,000

Real-world implementation gap and barriers

Quantified Implementation Deficit: Our analysis reveals a significant gap between research demonstration and practical deployment. The 18% real-world implementation rate in academic literature, expanding to 26% when including industry documentation, indicates substantial barriers to technology transfer.

Primary Implementation Barriers (Based on 23 Real-World Studies):

1. Data Quality and Infrastructure (78% of implementations):

- o Sensor calibration requirements exceed typical facility capabilities
- o Data integration complexity with legacy building management systems
- o Missing data handling affects 45% of deployments

2. Economic and Organizational Constraints (67% of implementations):

- o Upfront costs exceed budgetary expectations in 52% of attempts
- o Lack of skilled personnel for system maintenance and optimization
- o Organizational resistance to autonomous control systems

3. Technical Performance Issues (43% of implementations):

- o Real-world performance below simulation predictions by 15–25%
- o System reliability concerns during extreme weather conditions
- o Integration challenges with existing HVAC and lighting systems

Generalizability limitations and research bias

Building Type and Climate Bias: The predominant focus on office buildings (78%) and temperate climates (67%) limits generalizability to other contexts. Our subgroup analysis suggests that AI effectiveness varies significantly across building types, with educational facilities showing higher potential (24.8% average savings) but healthcare facilities demonstrating more modest gains (16.2%).

Geographic and Cultural Limitations:

- **Regional concentration:** 67% of studies from North America and Europe
- **Limited representation:** Developing countries represent < 8% of research
- **Cultural factors:** Occupant behavior patterns may not transfer across cultures

Methodological Limitations:

- **Simulation bias:** Controlled simulation environments may overestimate real-world performance
- **Publication bias:** Negative results likely underrepresented in academic literature
- **Temporal bias:** Short-term validation periods (median 6 months) insufficient for assessing long-term sustainability

Privacy, ethics, and regulatory compliance

Privacy and Data Protection Challenges: Current AI building systems collect extensive data about occupant behaviors, raising significant privacy concerns that are inadequately addressed in current research.

Regulatory Compliance Assessment:

- **GDPR Compliance:** Only 8% of studies explicitly address data anonymization
- **Building Codes:** Integration with existing ASHRAE and LEED standards unclear
- **Liability Issues:** Unclear responsibility allocation for AI system failures

Federated Learning and Privacy-Preserving Approaches: Emerging federated learning implementations show promise for maintaining data locality while enabling AI optimization. Analysis of 6 federated learning studies indicates 89% effectiveness retention compared to centralized approaches while addressing privacy concerns.

Algorithmic Fairness Considerations: Current optimization strategies risk producing biased comfort predictions or unequal energy distribution among different occupant groups. Only 3% of reviewed studies address fairness considerations in AI decision-making.

Long-term performance and maintenance strategies

Model Performance Degradation: Analysis of 8 multi-year studies reveals systematic performance degradation over time:

- **Year 1:** Baseline performance maintained
- **Year 2:** 8–15% performance decline without intervention
- **Year 3:** 15–25% performance decline without retraining

Maintenance Strategy Effectiveness:

- **Quarterly retraining:** Maintains 95% of initial performance
- **Semi-annual retraining:** Maintains 88% of initial performance
- **Annual retraining:** Maintains 78% of initial performance

Continuous Learning Implementations: Studies employing continuous learning algorithms ($n=12$) show superior long-term performance maintenance but require 34% higher computational resources and more sophisticated edge computing infrastructure.

Integration with smart grid and urban energy systems

Demand Response Potential: AI-optimized buildings demonstrate significant potential for grid services participation. Analysis of 5 grid-integrated studies suggests:

- **Demand response capacity:** 15–20% of building load available for grid services
- **Response time:** < 5 min for automated demand reduction
- **Revenue potential:** \$2.40–\$7.80 per kW-month in organized markets

Urban-Scale Coordination Benefits: Coordinated building AI systems could provide 15–20% of urban demand response capacity, contributing significantly to grid stability and renewable energy integration.

Table 5 Scalability evaluation matrix

Building category	Technical feasibility	Economic viability	Implementation complexity	Recommended approach
New construction (> 10,000 sq ft)	High	High	Medium	Full AI integration
Major renovation (> 5,000 sq ft)	High	Medium–High	Medium–High	Phased implementation
Existing buildings (> 5,000 sq ft)	Medium	Medium	High	Single-system pilots
Small buildings (< 5,000 sq ft)	Medium	Low–Medium	High	Focus on lighting/plug loads
Historic/protected buildings	Low	Low	Very High	Minimal intervention approaches

Synthesis of contributions, insights, and future research directions

Novel contributions and theoretical advances

First Comprehensive Taxonomic Framework: Our implementation-focused taxonomic framework represents a paradigm shift from purely technical categorization to deployment-centered classification. Unlike previous reviews that organized AI techniques by algorithmic characteristics, our framework prioritizes practical deployment considerations and implementation complexity.

Largest Meta-Analytic Synthesis: This study provides the first rigorous quantitative synthesis of AI building energy performance, analyzing 126 studies compared to previous maximum coverage of 45 studies. The meta-analysis reveals substantial performance heterogeneity ($I^2 = 45\text{--}81\%$) that previous qualitative reviews could not quantify.

Implementation Gap Quantification: We provide the first systematic assessment of the research-practice gap, revealing that only 18% of academic studies involve real-world deployment, expanding to 26% when including industry documentation. This quantification highlights a critical barrier to technology transfer.

Evidence-based performance benchmarks

Established Performance Hierarchy: Our meta-analysis establishes clear performance rankings with statistical significance:

1. **Hybrid Methods:** $28.1\% \pm 12.3\%$ energy savings (highest potential, highest variance)
2. **Reinforcement Learning:** $22.3\% \pm 8.4\%$ energy savings (strong performance, moderate variance)
3. **Supervised Learning:** $14.7\% \pm 5.2\%$ energy savings (consistent performance, lowest variance)

Economic Viability Thresholds: Analysis across building types establishes that AI optimization becomes economically viable for: (Table 5, Scalability Evaluation Matrix is given).

- Buildings larger than 5,000 sq ft in moderate climates
- Buildings larger than 2,000 sq ft in extreme climates
- Any building size when integrated with major renovation projects

Prioritized research agenda**Tier 1 Priorities (Immediate—1–2 years):****1. Multi-Season Real-World Validation Studies**

- o Target: 50 + buildings across diverse climate zones
- o Duration: Minimum 18-month validation periods
- o Focus: Performance consistency and degradation patterns

2. Standardized Implementation Framework Development

- o Sensor calibration protocols and data quality standards
- o Model validation and performance benchmarking procedures
- o Scalability assessment methodologies

3. Economic Modeling and Total Cost of Ownership Analysis

- o Comprehensive lifecycle cost analysis including maintenance
- o Regional economic models accounting for energy price variations
- o Risk assessment frameworks for implementation decisions

Tier 2 Priorities (Medium-term—3–5 years):**1. Occupant-Centric Optimization Research**

- o Personal comfort modeling and preference learning
- o Productivity impact assessment of environmental changes
- o Privacy-preserving personalization techniques

2. Multi-System Integration Studies

- o Coordinated HVAC, lighting, and renewable energy optimization
- o Building-to-grid integration and demand response capabilities
- o Urban-scale building coordination frameworks

3. Climate Adaptation and Resilience Research

- o Performance under extreme weather conditions
- o Adaptive algorithms for climate change scenarios
- o Integration with emergency response and disaster recovery

Tier 3 Priorities (Long-term—5 + years):**1. Advanced AI Architecture Development**

- o Federated learning for privacy-preserving optimization
- o Explainable AI for building operator trust and acceptance
- o Transfer learning for rapid deployment across building types

2. Policy and Regulatory Framework Development

- o Building code integration and compliance verification
- o Incentive structures for AI adoption
- o Liability and responsibility allocation frameworks

Practical implementation roadmap**Phase 1: Assessment and Planning (Months 1–3)**

- Building energy audit and baseline establishment
- Infrastructure readiness assessment (sensors, connectivity, computing)
- Stakeholder engagement and change management planning
- Economic feasibility analysis with site-specific parameters

Phase 2: Pilot Implementation (Months 4–9)

- Single-system deployment (typically HVAC or lighting)
- Basic sensor network installation and commissioning
- Initial algorithm training and performance validation
- User interface development and operator training

Phase 3: System Integration (Months 10–15)

- Multi-system coordination implementation
- Advanced sensor integration and edge computing deployment
- Continuous learning and adaptation mechanisms
- Performance monitoring and optimization refinement

Phase 4: Scaling and Optimization (Months 16–24)

- Full building integration and coordination
- Grid services integration where applicable
- Long-term performance monitoring and model maintenance
- Knowledge transfer and best practice documentation

Technology readiness and scalability assessment**Current Technology Readiness Levels:**

- **Supervised Learning Applications:** TRL 8–9 (proven technology, ready for deployment)
- **Reinforcement Learning Systems:** TRL 6–7 (technology demonstration, requires development for specific applications)
- **Hybrid Ensemble Methods:** TRL 5–6 (technology validation in relevant environment)
- **Edge AI Implementations:** TRL 4–5 (technology validation in laboratory environment)

Scalability Evaluation Matrix:**Risk Mitigation Strategies:**

Based on analysis of implementation challenges from real-world deployments:

1. Technical Risk Mitigation:

- o Comprehensive sensor network testing before AI deployment
- o Redundant communication pathways for critical systems
- o Manual override capabilities for all automated functions
- o Regular model performance monitoring and degradation detection

2. Economic Risk Mitigation:

- o Phased implementation with performance verification at each stage
- o Energy savings guarantees with vendor performance bonds
- o Total cost of ownership analysis including 10-year maintenance costs
- o Energy service company (ESCO) financing options for risk transfer

3. Organizational Risk Mitigation:

- o Comprehensive operator training and change management programs
- o Clear escalation procedures for system anomalies
- o Stakeholder engagement throughout implementation process
- o Performance dashboards for transparency and accountability

Implications, environmental impact, and future research directions

This systematic review and meta-analysis provides the most comprehensive assessment to date of artificial intelligence applications for building energy optimization, synthesizing evidence from 126 peer-reviewed studies spanning 2010–2024. The research establishes clear performance benchmarks, identifies critical implementation barriers, and provides evidence-based guidance for advancing AI deployment in smart buildings.

Key findings and evidence

Our meta-analysis demonstrates that AI-driven optimization achieves substantial energy savings with significant variation by approach: hybrid methods show highest potential ($28.1\% \pm 12.3\%$), reinforcement learning provides strong consistent performance ($22.3\% \pm 8.4\%$), and supervised learning offers reliable but modest gains ($14.7\% \pm 5.2\%$). These performance differences are both statistically significant ($p < 0.001$) and practically meaningful (effect sizes $d > 0.8$), providing clear guidance for technology selection.

Economic analysis reveals favorable return on investment across building contexts, with payback periods ranging from 2.1–5.8 years (median 3.4 years) and implementation costs of \$8,000–\$47,000 per facility. Economic viability improves significantly with building size and climate extremity, establishing clear deployment prioritization criteria.

Critical implementation insights

Despite promising technical performance, substantial barriers limit real-world deployment. Only 18% of academic studies report operational implementation, expanding to 26% when including industry documentation. Primary barriers include data quality requirements (78% of implementations), economic and organizational constraints (67%), and technical integration challenges (43%).

The research reveals concerning gaps in current approaches: 76% of studies focus on single-system optimization despite clear benefits of coordination, 83% validate performance over single seasonal cycles insufficient for long-term reliability assessment, and only 12% address critical privacy and ethical considerations.

Novel contributions

This study advances the field through several key contributions. We present the first implementation-focused taxonomic framework categorizing AI approaches by

deployment characteristics rather than purely technical features. Our meta-analysis provides the first rigorous quantitative synthesis with heterogeneity assessment and effect size analysis. We quantify the implementation gap between research and practice while identifying specific barriers limiting technology transfer.

Future research priorities

Based on systematic gap analysis, we establish a three-tier research agenda prioritizing immediate real-world validation studies, standardized implementation frameworks, and comprehensive economic modeling. Medium-term priorities include occupant-centric optimization research and multi-system integration studies, while long-term goals focus on advanced AI architectures and policy framework development.

Practical implications

For practitioners, this research provides evidence-based selection criteria for AI approaches based on building characteristics, climate conditions, and economic constraints. Our implementation roadmap offers concrete guidance for phased deployment strategies while our risk mitigation framework addresses common failure modes identified in real-world deployments.

Limitations and future work

Several limitations warrant acknowledgment. Geographic bias toward developed countries (92% of studies) and building type concentration on offices (78%) limit generalizability. Reliance on peer-reviewed literature may underestimate actual deployment rates due to industry publication patterns. Short validation periods in most studies (83% single-season) provide insufficient evidence for long-term performance sustainability.

Future research should prioritize multi-year validation studies across diverse building types and climate zones, development of standardized implementation protocols, and systematic documentation of industry deployments to bridge the research-practice gap.

Final assessment

This comprehensive analysis demonstrates that AI-driven building energy optimization has progressed beyond proof-of-concept to practical viability for appropriate applications. While significant implementation barriers remain, the combination of proven energy savings potential (22–28% for advanced approaches), favorable economics (median 3.4-year payback), and growing technological maturity supports accelerated deployment efforts.

Success requires moving beyond fragmented single-system approaches toward integrated, standardized implementations with appropriate attention to occupant needs, privacy concerns, and long-term performance sustainability. The evidence-based frameworks and prioritized research agenda presented here provide a foundation for this transition from research demonstration to widespread practical adoption.

Conclusion

His research presents a comprehensive analysis of the potential of artificial intelligence to optimize energy use in smart buildings, underpinned by an extensive meta-analysis of building energy consumption data and performance outcomes. The findings

demonstrate that AI methods—particularly reinforcement learning—can deliver significant energy savings while maintaining or enhancing occupant comfort.

The study contributes several key insights to the field. First, the meta-analysis indicates that AI-driven optimization achieves an average energy reduction of approximately 22%, corresponding to a potential 25% decrease in CO₂ emissions when scaled across building portfolios. These conclusions are grounded in robust statistical synthesis of data from diverse building types and operational conditions, highlighting the practical viability of AI systems under varying scenarios.

Beyond performance benchmarks, this research offers practical value by outlining frameworks for the implementation of AI-based energy management in commercial buildings. Cost–benefit analyses show favorable payback periods, confirming that AI-driven optimization is not only technically feasible but also economically attractive for building owners and operators.

The study also sets clear directions for future research and industry practice. Scaling up to pilot-scale real-world deployments and exploring integration with smart grid infrastructure are critical next steps. Further work is needed to develop standardized implementation guidelines, enhance data quality and interoperability, and ensure sustained system performance in complex, dynamic building environments.

Importantly, this review makes several novel contributions to the academic and practical understanding of AI in building energy management. It is the first systematic review to present a taxonomic framework categorizing AI techniques for this domain and provides the most comprehensive meta-analysis to date, covering 126 peer-reviewed studies. It also offers the first prioritized research agenda based explicitly on an analysis of implementation barriers, bridging the gap between theoretical advancements and practical deployment.

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Author contributions

Lakpriya Udayanga Gunasena Ekanayaka Gunasinghalge: Conceptualization, Data curation, Methodology, Software, Resource, Visualization, Formal Analysis, Supervision, Writing—original draft and review & editing. Ammar Alazab: Conceptualization, Methodology, Software, Resource, Supervision, Visualization, Formal Analysis, Writing—review & editing. Md. Alamin Talukdrer: Methodology, Data Curation, Visualization, Validation, Formal Analysis, and Writing (Review & Editing).

Data availability

No datasets were generated or analysed during the current study.

Declarations

Conflict of interest

The authors declare no competing interests.

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